NEOGÁS

AR CONDICIONADO

MEMÓRIA DE CÁLCULO

RUA SÃO JOSÉ, Nº 70 - 13º ANDAR,
CENTRO, RIO DE JANEIRO - RJ
FEVEREIRO / 2024

MAIA
consultoria e projetos de ar condicionado

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ENGENHARIA DE AR-CONDICIONADO

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MAIA consultoria e projetos de ar condicionado

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NEOGÁS – UNIDADE CENTRO, RIO DE JANEIRO - RJ

1. OBJETIVO

O presente documento tem por objetivo apresentar a memória de cálculo das cargas térmicas do sistema de ar-condicionado do 13º Pavimento da unidade NEOGÁS localizada na Rua São José, Nº 70 - 13º andar, Centro - Rio de Janeiro - RJ.

Para o cálculo dos requisitos de ar exterior e exaustão, foram utilizadas planilhas do aplicativo Microsoft Excel. Já para o cálculo da carga térmica, foram empregadas planilhas do Excel em conjunto com o programa *'Hourly Analysis Program (HAP)'*, versão 4.90 da Carrier Corporation.

2. NORMAS APLICÁVEIS

Todos os cálculos envolvidos, dimensionamento, especificações, bem como fornecimento e instalação, devem obedecer às normas, regulamentos e publicações técnicas a seguir:

- NBR 15220 – Desempenho térmico de edificações;
- NBR-16401-1 – Instalações de ar-condicionado – Sistemas centrais e unitários – Parte 1 – Projetos das instalações;
- NBR-16401-2 – Instalações de ar-condicionado – Sistemas centrais e unitários – Parte 2 – Parâmetros de conforto térmico;
- NBR-16401-3 – Instalações de ar-condicionado – Sistemas centrais e unitários – Parte 3 – Qualidade do ar interior.
- NBR-10151 – Acústica-Avaliação do Ruído em áreas habitadas.
- NBR-5410 – Instalações elétricas baixas tensão.
- NR (Normas Regulamentares) – Ministério do Trabalho;
- Regulamento GM/MS nº3523/1998 – Ministério da Saúde;
- Resolução RE-09: 2003 da ANVISA;
- Decreto n.º 22281 de 19 de novembro 2002 - Regulamento para a Instalação e Conservação de Sistemas de Ar-Condicionado e Ventilação Mecânica no Município do Rio de Janeiro.
- AMCA – “Air Movement and Control Association”.
- Publicações técnicas da ASHRAE – “American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc”.
- SMACNA – “Sheet Metal and Air Conditioning Contractors National Association”.

3. DADOS DE ENTRADA

Este capítulo apresenta uma descrição detalhada dos parâmetros de entrada utilizados no projeto. Isso inclui informações climáticas específicas do local do projeto, dados psicométricos relevantes, fator de segurança adotado e outros parâmetros essenciais utilizados no cálculo da carga térmica.

3.1 LOCALIZAÇÃO

Cidade: Rio de Janeiro

Referência: Aeroporto do Galeão

Altitude: 6 m

Latitude: 22,82º Sul

Longitude: 43,25º Oeste

3.2 CONDIÇÕES PSICROMÉTRICAS

3.2.1 Dados da norma ABNT NBR 16401-3:2008

Segundo a norma ABNT NBR 16401-3:2008, as condições psicrométricas do ar externo para o percentil 0,4 dos dias mais quentes do ano e para o percentil 99,6 dos dias mais frios do ano são:

3.2.1.1 RESFRIAMENTO:

Referência: Aeroporto Galeão - RJ

Mês de maior temperatura: fevereiro

TBS Máximo: 38,1 ºC

TBU Coincidente: 25,6 ºC

Média mensal da variação de temperatura diária (ΔT): 9,8 ºC

3.2.2 Condições do ar interno refrigerado

A carga térmica será calculada com o objetivo de manter o ar interno refrigerado no intervalo de temperatura a seguir:

Ar Interno (TBS/UR)[1]: 22,0 °C ± 1,0 °C

[1] Temperatura de conforto conforme recomendado na seção 5 da norma ABNT NBR 16401-2:2008.

3.3 COEFICIENTES DE SEGURANÇA ADOTADOS

Calor sensível: 10%

Calor latente: 10%

3.4 FATOR DE BY-PASS DAS SERPENTINAS

Foi considerado fator de by-pass das serpentinas de 15%

3.5 NÍVEL DE RUÍDO

Os níveis de ruído permitidos em cada área foram determinados com base nas recomendações das normas ABNT NBR-16401-1, item 7.4 e ABNT NBR-10152, tabela I. Além disso, essas normas estão em conformidade com as normas técnicas da ASHRAE e SMACNA. As normas que seguimos são listadas a seguir:

ABNT NBR 16401

ABNT NBR 10152

ASHRAE Handbook – Application – 2019 – Capítulo 49

ASHRAE Handbook – Fundamentals – 2017 – Capítulo 8

SMACNA – 1990 – Chapter 11 – HVAC Systems duct design.

3.6 COEFICIENTES DE TRANSMISSÃO

Parede interna: 2,54 W/m² ºC

Parede externa: 2,31 W/m² ºC (coeficiente de sombreamento de 90%)

Laje interna: 2,20 W/m² ºC

Vidro Simples: 5,9 W/m² ºC (coeficiente de sombreamento de 0,7)

3.7 DISSIPAÇÃO TÉRMICA DE ILUMINAÇÃO E EQUIPAMENTOS

Os valores adotados para a carga térmica de iluminação e equipamentos foram baseados nas recomendações da NBR 16401-1, em conjunto com a análise da planta baixa de arquitetura fornecida. Para os equipamentos que não são especificados nas normas, recorremos a valores padrão de equipamentos semelhantes quando a especificação está disponível, ou com base em similares disponíveis no mercado nacional.

3.8 DISSIPAÇÃO TÉRMICA DEVIDO A PRESENÇA HUMANA

Sentado, trabalho leve:

Carga sensível: 70 W/pessoa (NBR 16401-1)

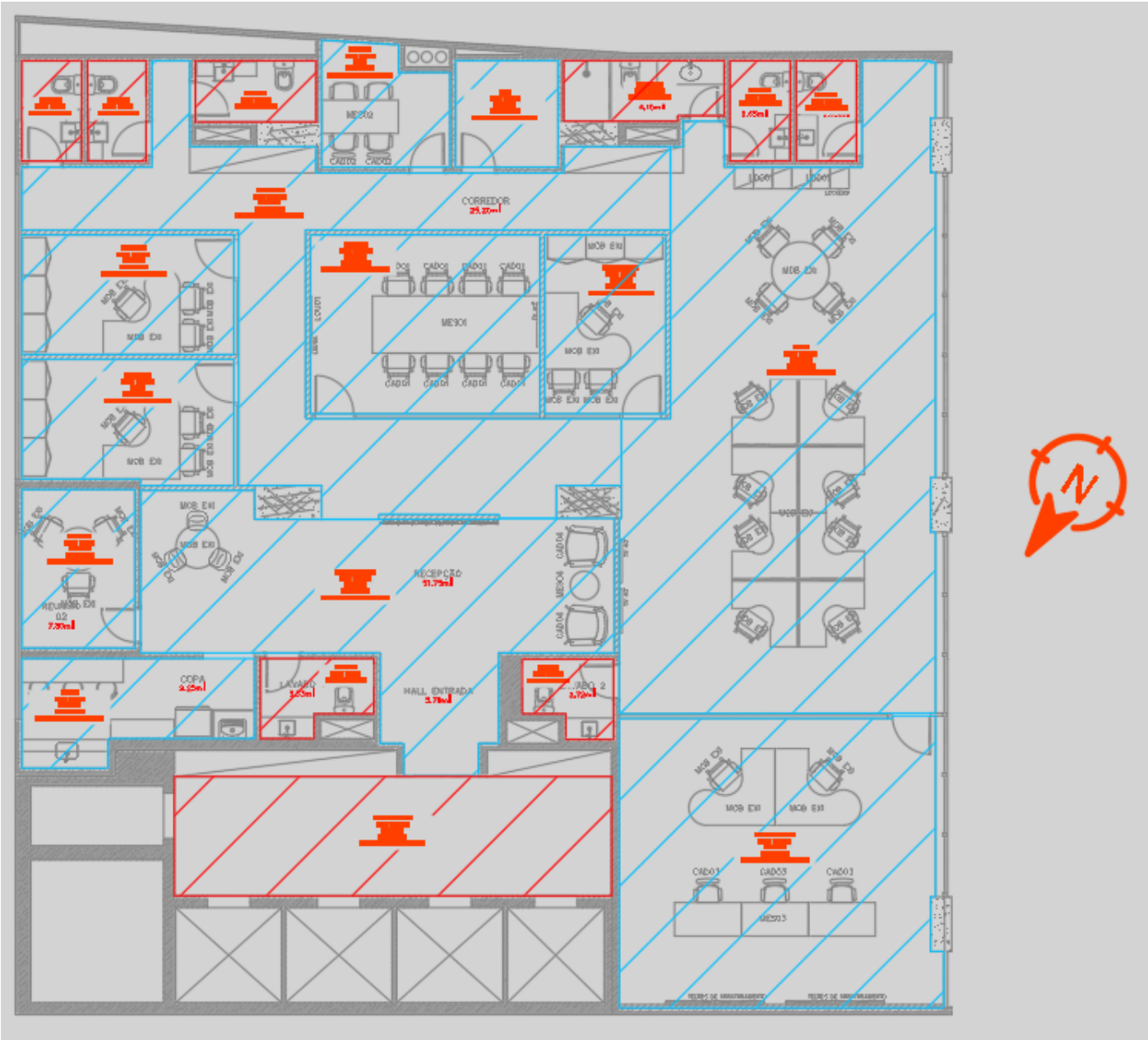
Carga latente: 45 W/pessoa (NBR 16401-1)

Parado em pé, trabalho moderado; caminhando:

Carga sensível: 75 W/pessoa (NBR 16401-1)

Carga latente: 55 W/pessoa (NBR 16401-1)

4. ZONEAMENTO DAS ZONAS TÉRMICAS



LEGENDA DE ZONAS TÉRMICAS - HVAC

- ZONA TÉRMICA DE INSUFLAMENTO DE AR CLIMATIZADO (ANSI 31 HATCH)
- ZONA TÉRMICA DE EXAUSTÃO (ANSI 31 HATCH)
- ZONA TÉRMICA DE FLUXO DE AR EXTERIOR (ANSI 31 HATCH)
- ZONA SEM CLIMATIZAÇÃO E/OU SEM INTERVENÇÃO (ANSI 31 HATCH)

5. VAZÃO DE AR EXTERIOR

As taxas de ar exterior foram calculadas conforme requerimentos da ANVISA e ABNT NBR 16401-3:2008 (NÍVEL 2)

Ambientes	Área	Pessoas	Requisito por pessoa ANVISA	Requisito por pessoa ABNT	Requisito por área ABNT	Vazão de Ar Exterior MÍNIMA
Corredor 1	40.98 m^2	7 Pessoas	7.50 L/s por pessoa	5.30 L/s por pessoa	0.40 L/s por m^2	192.6 m^3/h
Reunião 3	6.31 m^2	4 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	108.0 m^3/h
CPD 1	4.78 m^2	1 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	27.0 m^3/h
Staff 1	73.90 m^2	12 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	324.0 m^3/h
Diretória 1	11.07 m^2	3 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	81.0 m^3/h
Diretória 2	11.36 m^2	3 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	81.0 m^3/h
Reunião 1	17.95 m^2	8 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	216.0 m^3/h
Gerência SSMA	9.46 m^2	3 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	81.0 m^3/h
Reunião 2	7.83 m^2	3 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	81.0 m^3/h
Recepção e hall	34.86 m^2	5 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	135.0 m^3/h
Monitoramento	40.14 m^2	5 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	135.0 m^3/h
Copa 1	9.24 m^2	4 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	1.10 L/s por m^2	108.0 m^3/h
Total						1650.6 m^3/h

6. CARGA TÉRMICA

6.1 RESUMO DOS VALORES DE CARGAS TÉRMICAS

Ambiente	Vazão de ar insuflado	Vazão de ar insuflado	Carga térmica em kW	Carga térmica em TR	Carga térmica em BTU/h
Corredor 1	337 L/s	1213 m^3/h	7.60 kW	2.16 TR	25 932 BTU/h
CPD 1	217 L/s	781 m^3/h	2.80 kW	0.80 TR	9 554 BTU/h
Diretória 1	139 L/s	500 m^3/h	2.40 kW	0.68 TR	8 189 BTU/h
Diretória 2	139 L/s	500 m^3/h	2.40 kW	0.68 TR	8 189 BTU/h
Gerência SSMA	37 L/s	133 m^3/h	1.40 kW	0.40 TR	4 777 BTU/h
Monitoramento	446 L/s	1606 m^3/h	6.90 kW	1.96 TR	23 544 BTU/h
Recepção e Hall	328 L/s	1181 m^3/h	7.00 kW	1.99 TR	23 885 BTU/h
Reunião 1	136 L/s	490 m^3/h	4.20 kW	1.19 TR	14 331 BTU/h
Reunião 2	168 L/s	605 m^3/h	2.80 kW	0.80 TR	9 554 BTU/h
Reunião 3	96 L/s	346 m^3/h	2.40 kW	0.68 TR	8 189 BTU/h
Staff 1	1010 L/s	3636 m^3/h	18.50 kW	5.26 TR	63 125 BTU/h
Copa	227 L/s	817 m^3/h	3.9 kW	1.11 TR	13 307 BTU/h

Air System Sizing Summary for AC-01

Project Name: NEOGÁS - 13º

02/25/2024

1. AIR SYSTEM INFORMATION

Air System Name	AC-01	Number of zones	1
Equipment Class	SPLT AHU	Floor Area	41,0 m²
Air System Type	SZCAV	Location	Rio de Janeiro, Brazil

2. Sizing Calculation Information

Calculation Months	Jan to Dec	Zone L/s Sizing	Sum of space airflow rates
Sizing Data	Calculated	Space L/s Sizing	Individual peak space loads

3. Central Cooling Coil Sizing Data

Total coil load	7,6	kW	Load occurs at	Jun 1500
Sensible coil load	4,8	kW	OA DB / WB	37,5 / 25,6 °C
Coil L/s at Jun 1500	337	L/s	Entering DB / WB	25,2 / 19,4 °C
Max block L/s	337	L/s	Leaving DB / WB	13,5 / 12,8 °C
Sum of peak zone L/s	337	L/s	Coil ADP	11,5 °C
Sensible heat ratio	0,630		Bypass Factor	0,150
m²/kW	5,4		Resulting RH	64 %
W/m²	184,4		Design supply temp.	13,0 °C
Water flow @ 5,6 °K rise	N/A		Zone T-stat Check	1 of 1 OK
			Max zone temperature deviation	0,0 °K

4. Supply Fan Sizing Data

Actual max L/s	337	L/s	Fan motor BHP	0,24	BHP
Standard L/s	337	L/s	Fan motor kW	0,19	kW
Actual max L/(s-m²)	8,22	L/(s-m²)	Fan static	300	Pa

5. Outdoor Ventilation Air Data

Design airflow L/s	54	L/s	L/s/person	7,64	L/s/person
L/(s-m²)	1,30	L/(s-m²)			

6. Air System Information

Air System Name **AC-01**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **41,0** m²
Location **Rio de Janeiro, Brazil**

7. Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

8. Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	3,6	330	330	Aug 1500	0,1	41,0	8,06

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Corredor 1	1	3,6	Aug 1500	330	0,1	41,0	8,06

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jun 1500 COOLING OA DB / WB 37,5 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	2 m²	121	-	2 m²	-	-
Wall Transmission	2 m²	68	-	2 m²	19	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	2 m²	116	-	2 m²	42	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	492 W	268	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	250 W	225	-	0	0	-
People	7	379	385	0	0	0
Infiltration	-	1969	1472	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	315	186	10%	6	0
>> Total Zone Loads	**-**	**3460**	**2042**	**-**	**67**	**0**
Zone Conditioning	-	3379	2042	-	-70	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	148	-	0	0	-
Return Fan Load	337 L/s	0	-	337 L/s	0	-
Ventilation Load	54 L/s	944	756	54 L/s	262	0
Supply Fan Load	337 L/s	187	-	337 L/s	-187	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	104	-	3%	2	-
>> Total System Loads	**-**	**4761**	**2799**	**-**	**7**	**0**
Central Cooling Coil	-	4761	2799	-	0	0
>> Total Conditioning	**-**	**4761**	**2799**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

June DESIGN COOLING DAY, 1500

9. TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	37,5	0,01578	54	400	944	756
Vent - Return Mixing	Outlet	25,2	0,01174	337	621	-	-
Central Cooling Coil	Outlet	13,5	0,00892	337	621	4761	2799
Supply Fan	Outlet	14,0	0,00892	337	621	187	-
Cold Supply Duct	Outlet	14,3	0,00892	330	621	-	-
Zone Air	-	22,7	0,01102	330	663	3379	2042
Return Plenum	Outlet	23,1	0,01102	330	663	148	-
Duct Leakage Air	Outlet	14,0	0,00892	7	621	-	-
Return Duct	Outlet	22,9	0,01098	337	662	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s)
Site Altitude = 6,0 m

10. TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	3460	Cooling	3379	22,7	330	663	0	0

Air System Information

Air System Name	**AC-02**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**4,8** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2,8**	kW
Sensible coil load	**2,6**	kW
Coil L/s at Jul 1500	**217**	L/s
Max block L/s	**217**	L/s
Sum of peak zone L/s	**217**	L/s
Sensible heat ratio	**0,924**	
m²/kW	**1,7**	
W/m²	**592,1**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Jul 1500**	
OA DB / WB	**38,1 / 25,6**	°C
Entering DB / WB	**23,2 / 16,3**	°C
Leaving DB / WB	**13,2 / 12,2**	°C
Coil ADP	**11,4**	°C
Bypass Factor	**0,150**	
Resulting RH	**49**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**217**	L/s
Standard L/s	**217**	L/s
Actual max L/(s-m²)	**45,23**	L/(s-m²)
Fan motor BHP	**0,15**	BHP
Fan motor kW	**0,12**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**8**	L/s
L/(s-m²)	**1,56**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-02

Air System Information

Air System Name **AC-02**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **4,8** m²
Location **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	2,3	213	213	Oct 1500	0,1	4,8	44,33

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
CPD 1	1	2,3	Oct 1500	213	0,1	4,8	44,33

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 38,1 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	5 m²	194	-	5 m²	62	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	58 W	31	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	2000 W	1801	-	0	0	-
People	1	54	55	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	208	6	10%	6	0
>> Total Zone Loads	**-**	**2289**	**61**	**-**	**68**	**0**
Zone Conditioning	-	2280	61	-	65	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	17	-	0	0	-
Return Fan Load	217 L/s	0	-	217 L/s	0	-
Ventilation Load	8 L/s	140	155	8 L/s	43	0
Supply Fan Load	217 L/s	121	-	217 L/s	-121	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	69	-	3%	2	-
>> Total System Loads	**-**	**2626**	**215**	**-**	**-11**	**0**
Central Cooling Coil	-	2626	216	-	0	0
>> Total Conditioning	**-**	**2626**	**216**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	38,1	0,01554	8	400	140	155
Vent - Return Mixing	Outlet	23,2	0,00878	217	842	-	-
Central Cooling Coil	Outlet	13,2	0,00844	217	842	2626	216
Supply Fan	Outlet	13,6	0,00844	217	842	121	-
Cold Supply Duct	Outlet	13,9	0,00844	213	842	-	-
Zone Air	-	22,8	0,00854	213	858	2280	61
Return Plenum	Outlet	22,8	0,00854	213	858	17	-
Duct Leakage Air	Outlet	13,6	0,00844	4	842	-	-
Return Duct	Outlet	22,7	0,00854	217	858	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	2289	Cooling	2280	22,8	213	858	0	0

Air System Information

Air System Name	**AC-03**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**11,1** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2,4**	kW
Sensible coil load	**1,9**	kW
Coil L/s at Oct 1400	**139**	L/s
Max block L/s	**139**	L/s
Sum of peak zone L/s	**139**	L/s
Sensible heat ratio	**0,785**	
m²/kW	**4,6**	
W/m²	**218,1**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Oct 1400**	
OA DB / WB	**35,6 / 24,4**	°C
Entering DB / WB	**24,8 / 17,9**	°C
Leaving DB / WB	**13,5 / 12,5**	°C
Coil ADP	**11,5**	°C
Bypass Factor	**0,150**	
Resulting RH	**53**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**139**	L/s
Standard L/s	**139**	L/s
Actual max L/(s-m²)	**12,54**	L/(s-m²)
Fan motor BHP	**0,10**	BHP
Fan motor kW	**0,08**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**23**	L/s
L/(s-m²)	**2,03**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-03

Air System Information

Air System Name **AC-03**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **11,1** m²
Location **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,5	136	136	Dec 1200	0,1	11,1	12,29

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Diretória 1	1	1,5	Dec 1200	136	0,1	11,1	12,29

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Oct 1400 COOLING OA DB / WB 35,6 °C / 24,4 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	4 m²	532	-	4 m²	-	-
Wall Transmission	2 m²	117	-	2 m²	27	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	4 m²	248	-	4 m²	105	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	133 W	71	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	163 W	146	-	0	0	-
People	3	147	135	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	126	14	10%	13	0
>> Total Zone Loads	**-**	**1388**	**149**	**-**	**145**	**0**
Zone Conditioning	-	1392	149	-	38	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	40	-	0	0	-
Return Fan Load	139 L/s	0	-	139 L/s	0	-
Ventilation Load	23 L/s	349	373	23 L/s	32	0
Supply Fan Load	139 L/s	77	-	139 L/s	-77	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	42	-	3%	4	-
>> Total System Loads	**-**	**1900**	**522**	**-**	**-2**	**0**
Central Cooling Coil	-	1900	521	-	0	0
>> Total Conditioning	**-**	**1900**	**521**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

October DESIGN COOLING DAY, 1400

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	35,6	0,01465	23	400	349	373
Vent - Return Mixing	Outlet	24,8	0,00993	139	908	-	-
Central Cooling Coil	Outlet	13,5	0,00866	139	908	1900	521
Supply Fan	Outlet	13,9	0,00866	139	908	77	-
Cold Supply Duct	Outlet	14,2	0,00866	136	908	-	-
Zone Air	-	22,6	0,00903	136	1008	1392	149
Return Plenum	Outlet	22,9	0,00903	136	1008	40	-
Duct Leakage Air	Outlet	13,9	0,00866	3	908	-	-
Return Duct	Outlet	22,7	0,00902	139	1006	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1388	Cooling	1392	22,6	136	1008	0	0

Air System Information

Air System Name	**AC-04**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**11,4** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2,4**	kW
Sensible coil load	**1,9**	kW
Coil L/s at Sep 1500	**139**	L/s
Max block L/s	**139**	L/s
Sum of peak zone L/s	**139**	L/s
Sensible heat ratio	**0,779**	
m²/kW	**4,7**	
W/m²	**212,0**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Sep 1500**	
OA DB / WB	**37,0 / 25,0**	°C
Entering DB / WB	**25,1 / 18,2**	°C
Leaving DB / WB	**13,9 / 12,9**	°C
Coil ADP	**11,9**	°C
Bypass Factor	**0,150**	
Resulting RH	**54**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**139**	L/s
Standard L/s	**139**	L/s
Actual max L/(s-m²)	**12,23**	L/(s-m²)
Fan motor BHP	**0,10**	BHP
Fan motor kW	**0,08**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**23**	L/s
L/(s-m²)	**1,97**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-04

Air System Information

Air System Name **AC-04**
Equipment Class ..... **SPLT AHU**
Air System Type ..... **SZCAV**

Number of zones **1**
Floor Area ..... **11,4** m²
Location ..... **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months ..... **Jan to Dec**
Sizing Data ..... **Calculated**

Zone L/s Sizing ..... **Sum of space airflow rates**
Space L/s Sizing ..... **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,5	137	137	Dec 1200	0,1	11,4	11,98

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Diretória 2	1	1,5	Dec 1200	137	0,1	11,4	11,98

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Sep 1500 COOLING OA DB / WB 37,0 °C / 25,0 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	4 m²	448	-	4 m²	-	-
Wall Transmission	2 m²	118	-	2 m²	27	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	4 m²	278	-	4 m²	105	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	137 W	74	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	163 W	147	-	0	0	-
People	3	151	135	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	122	14	10%	13	0
>> Total Zone Loads	**-**	**1338**	**149**	**-**	**145**	**0**
Zone Conditioning	-	1338	149	-	38	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	41	-	0	0	-
Return Fan Load	139 L/s	0	-	139 L/s	0	-
Ventilation Load	23 L/s	385	385	23 L/s	33	0
Supply Fan Load	139 L/s	77	-	139 L/s	-77	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	40	-	3%	4	-
>> Total System Loads	**-**	**1882**	**534**	**-**	**-2**	**0**
Central Cooling Coil	-	1882	534	-	0	0
>> Total Conditioning	**-**	**1882**	**534**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

September DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	37,0	0,01509	23	400	385	385
Vent - Return Mixing	Outlet	25,1	0,01021	139	908	-	-
Central Cooling Coil	Outlet	13,9	0,00891	139	908	1882	534
Supply Fan	Outlet	14,3	0,00891	139	908	77	-
Cold Supply Duct	Outlet	14,6	0,00891	137	908	-	-
Zone Air	-	22,7	0,00928	137	1008	1338	149
Return Plenum	Outlet	23,0	0,00928	137	1008	41	-
Duct Leakage Air	Outlet	14,3	0,00891	3	908	-	-
Return Duct	Outlet	22,8	0,00927	139	1005	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1338	Cooling	1338	22,7	137	1008	0	0

Air System Information

Air System Name	**AC-05**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**9,5** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1,4**	kW
Sensible coil load	**0,9**	kW
Coil L/s at Jul 1500	**37**	L/s
Max block L/s	**37**	L/s
Sum of peak zone L/s	**37**	L/s
Sensible heat ratio	**0,617**	
m²/kW	**6,8**	
W/m²	**146,2**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Jul 1500**	
OA DB / WB	**38,1 / 25,6**	°C
Entering DB / WB	**32,3 / 22,6**	°C
Leaving DB / WB	**13,2 / 12,2**	°C
Coil ADP	**9,8**	°C
Bypass Factor	**0,150**	
Resulting RH	**56**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**37**	L/s
Standard L/s	**37**	L/s
Actual max L/(s-m²)	**3,92**	L/(s-m²)
Fan motor BHP	**0,03**	BHP
Fan motor kW	**0,02**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**23**	L/s
L/(s-m²)	**2,37**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-05

Air System Information

Air System Name: **AC-05**
Equipment Class: **SPLT AHU**
Air System Type: **SZCAV**

Number of zones: **1**
Floor Area: **9,5** m²
Location: **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months: **Jan to Dec**
Sizing Data: **Calculated**

Zone L/s Sizing: **Sum of space airflow rates**
Space L/s Sizing: **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	0,4	36	36	Jan 1500	0,0	9,5	3,84

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Gerência SSMA	1	0,4	Jan 1500	36	0,0	9,5	3,84

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 38,1 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	114 W	62	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	163 W	147	-	0	0	-
People	3	151	135	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	36	14	10%	0	0
>> Total Zone Loads	**-**	**396**	**149**	**-**	**0**	**0**
Zone Conditioning	-	391	149	-	-11	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	34	-	0	0	-
Return Fan Load	37 L/s	0	-	37 L/s	0	-
Ventilation Load	23 L/s	399	383	23 L/s	32	0
Supply Fan Load	37 L/s	21	-	37 L/s	-21	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	12	-	3%	0	-
>> Total System Loads	**-**	**857**	**532**	**-**	**0**	**0**
Central Cooling Coil	-	857	532	-	0	0
>> Total Conditioning	**-**	**857**	**532**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	38,1	0,01554	23	400	399	383
Vent - Return Mixing	Outlet	32,3	0,01326	37	664	-	-
Central Cooling Coil	Outlet	13,2	0,00841	37	664	857	532
Supply Fan	Outlet	13,7	0,00841	37	664	21	-
Cold Supply Duct	Outlet	13,9	0,00841	36	664	-	-
Zone Air	-	22,8	0,00979	36	1075	391	149
Return Plenum	Outlet	23,6	0,00979	36	1075	34	-
Duct Leakage Air	Outlet	13,7	0,00841	1	664	-	-
Return Duct	Outlet	23,4	0,00976	37	1067	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	396	Cooling	391	22,8	36	1075	0	0

Air System Information

Air System Name	**AC-06**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**40,1** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**6,9**	kW
Sensible coil load	**5,9**	kW
Coil L/s at Jun 1500	**446**	L/s
Max block L/s	**446**	L/s
Sum of peak zone L/s	**446**	L/s
Sensible heat ratio	**0,850**	
m²/kW	**5,8**	
W/m²	**172,6**	
Water flow @ 5,6 °K rise	**N/A**	

Load occurs at	**Jun 1500**	
OA DB / WB	**37,5 / 25,6**	°C
Entering DB / WB	**24,1 / 17,0**	°C
Leaving DB / WB	**13,1 / 12,2**	°C
Coil ADP	**11,2**	°C
Bypass Factor	**0,150**	
Resulting RH	**50**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**446**	L/s
Standard L/s	**445**	L/s
Actual max L/(s-m²)	**11,11**	L/(s-m²)

Fan motor BHP	**0,31**	BHP
Fan motor kW	**0,25**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**38**	L/s
L/(s-m²)	**0,94**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-06

Air System Information

Air System Name **AC-06**
Equipment Class ..... **SPLT AHU**
Air System Type ..... **SZCAV**

Number of zones **1**
Floor Area ..... **40,1** m²
Location ..... **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months ..... **Jan to Dec**
Sizing Data ..... **Calculated**

Zone L/s Sizing ..... **Sum of space airflow rates**
Space L/s Sizing ..... **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	4,7	437	437	Jun 1600	0,5	40,1	10,89

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Monitoramento	1	4,7	Jun 1600	437	0,5	40,1	10,89

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jun 1500 COOLING OA DB / WB 37,5 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	6 m²	937	-	6 m²	-	-
Wall Transmission	25 m²	1123	-	25 m²	293	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	6 m²	463	-	6 m²	168	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	481 W	262	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	1380 W	1243	-	0	0	-
People	5	252	225	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	428	23	10%	46	0
>> Total Zone Loads	**-**	**4708**	**248**	**-**	**507**	**0**
Zone Conditioning	-	4682	248	-	153	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	144	-	0	0	-
Return Fan Load	446 L/s	0	-	446 L/s	0	-
Ventilation Load	38 L/s	666	792	38 L/s	69	0
Supply Fan Load	446 L/s	248	-	446 L/s	-248	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	141	-	3%	15	-
>> Total System Loads	**-**	**5880**	**1040**	**-**	**-10**	**0**
Central Cooling Coil	-	5880	1040	-	0	0
>> Total Conditioning	**-**	**5880**	**1040**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

June DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	37,5	0,01578	38	400	666	792
Vent - Return Mixing	Outlet	24,1	0,00921	446	902	-	-
Central Cooling Coil	Outlet	13,1	0,00842	446	902	5880	1040
Supply Fan	Outlet	13,6	0,00842	446	902	248	-
Cold Supply Duct	Outlet	13,9	0,00842	437	902	-	-
Zone Air	-	22,7	0,00861	437	950	4682	248
Return Plenum	Outlet	23,0	0,00861	437	950	144	-
Duct Leakage Air	Outlet	13,6	0,00842	9	902	-	-
Return Duct	Outlet	22,8	0,00860	446	949	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	4708	Cooling	4682	22,7	437	950	0	0

Air System Information

Air System Name	**AC-07**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**34,9** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**7,0**	kW
Sensible coil load	**4,4**	kW
Coil L/s at Jul 1500	**328**	L/s
Max block L/s	**328**	L/s
Sum of peak zone L/s	**328**	L/s
Sensible heat ratio	**0,632**	
m²/kW	**5,0**	
W/m²	**201,0**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Jul 1500**	
OA DB / WB	**38,1 / 25,6**	°C
Entering DB / WB	**24,7 / 19,1**	°C
Leaving DB / WB	**13,5 / 12,8**	°C
Coil ADP	**11,5**	°C
Bypass Factor	**0,150**	
Resulting RH	**64**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**328**	L/s
Standard L/s	**328**	L/s
Actual max L/(s-m²)	**9,40**	L/(s-m²)
Fan motor BHP	**0,23**	BHP
Fan motor kW	**0,18**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**38**	L/s
L/(s-m²)	**1,07**	L/(s-m²)
L/s/person	**9,38**	L/s/person

Zone Sizing Summary for AC-07

Air System Information

Air System Name **AC-07**
Equipment Class ..... **SPLT AHU**
Air System Type ..... **SZCAV**

Number of zones **1**
Floor Area ..... **34,9** m²
Location ..... **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months ..... **Jan to Dec**
Sizing Data ..... **Calculated**

Zone L/s Sizing ..... **Sum of space airflow rates**
Space L/s Sizing ..... **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	3,5	322	322	Jul 1500	0,0	34,9	9,22

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Recepção e Hall	1	3,5	Jul 1500	322	0,0	34,9	9,22

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 38,1 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	419 W	228	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	250 W	225	-	0	0	-
People	4	216	220	0	0	0
Infiltration	-	2505	1677	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	317	190	10%	0	0
>> Total Zone Loads	**-**	**3492**	**2087**	**-**	**0**	**0**
Zone Conditioning	-	3337	2087	-	-26	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	126	-	0	0	-
Return Fan Load	328 L/s	0	-	328 L/s	0	-
Ventilation Load	38 L/s	684	492	38 L/s	219	0
Supply Fan Load	328 L/s	182	-	328 L/s	-182	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	105	-	3%	0	-
>> Total System Loads	**-**	**4434**	**2579**	**-**	**10**	**0**
Central Cooling Coil	-	4434	2579	-	0	0
>> Total Conditioning	**-**	**4434**	**2579**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	38,1	0,01554	38	400	684	492
Vent - Return Mixing	Outlet	24,7	0,01160	328	531	-	-
Central Cooling Coil	Outlet	13,5	0,00893	328	531	4434	2579
Supply Fan	Outlet	14,0	0,00893	328	531	182	-
Cold Supply Duct	Outlet	14,2	0,00893	322	531	-	-
Zone Air	-	22,8	0,01113	322	548	3337	2087
Return Plenum	Outlet	23,2	0,01113	322	548	126	-
Duct Leakage Air	Outlet	14,0	0,00893	7	531	-	-
Return Duct	Outlet	23,0	0,01109	328	547	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	3492	Cooling	3337	22,8	322	548	0	0

Air System Information

Air System Name	AC-08	Number of zones	1
Equipment Class	SPLT AHU	Floor Area	18,0 m²
Air System Type	SZCAV	Location	Rio de Janeiro, Brazil

Sizing Calculation Information

Calculation Months	Jan to Dec		
Sizing Data	Calculated	Zone L/s Sizing	Sum of space airflow rates
		Space L/s Sizing	Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load	4,2	kW	Load occurs at	Jul 1400
Sensible coil load	2,7	kW	OA DB / WB	37,8 / 25,5 °C
Coil L/s at Jul 1400	136	L/s	Entering DB / WB	29,5 / 21,0 °C
Max block L/s	136	L/s	Leaving DB / WB	13,1 / 12,1 °C
Sum of peak zone L/s	136	L/s	Coil ADP	10,2 °C
Sensible heat ratio	0,644		Bypass Factor	0,150
m²/kW	4,3		Resulting RH	55 %
W/m²	232,2		Design supply temp.	13,0 °C
Water flow @ 5,6 °K rise	N/A		Zone T-stat Check	1 of 1 OK
			Max zone temperature deviation	0,0 °K

Supply Fan Sizing Data

Actual max L/s	136	L/s	Fan motor BHP	0,10	BHP
Standard L/s	136	L/s	Fan motor kW	0,08	kW
Actual max L/(s-m²)	7,57	L/(s-m²)	Fan static	300	Pa

Outdoor Ventilation Air Data

Design airflow L/s	60	L/s	L/s/person	7,50	L/s/person
L/(s-m²)	3,33	L/(s-m²)			

Zone Sizing Summary for AC-08

Air System Information

Air System Name **AC-08**
Equipment Class ..... **SPLT AHU**
Air System Type ..... **SZCAV**

Number of zones **1**
Floor Area ..... **18,0** m²
Location ..... **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months ..... **Jan to Dec**
Sizing Data ..... **Calculated**

Zone L/s Sizing ..... **Sum of space airflow rates**
Space L/s Sizing ..... **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,4	133	133	Jan 1500	0,0	18,0	7,41

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Reunião 1	1	1,4	Jan 1500	133	0,0	18,0	7,41

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1400 COOLING OA DB / WB 37,8 °C / 25,5 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	216 W	115	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	884 W	790	-	0	0	-
People	8	393	360	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	130	36	10%	0	0
>> Total Zone Loads	**-**	**1427**	**396**	**-**	**0**	**0**
Zone Conditioning	-	1433	396	-	-12	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	65	-	0	0	-
Return Fan Load	136 L/s	0	-	136 L/s	0	-
Ventilation Load	60 L/s	1077	1091	60 L/s	92	0
Supply Fan Load	136 L/s	76	-	136 L/s	-76	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	43	-	3%	0	-
>> Total System Loads	**-**	**2693**	**1487**	**-**	**4**	**0**
Central Cooling Coil	-	2693	1487	-	0	0
>> Total Conditioning	**-**	**2693**	**1487**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

July DESIGN COOLING DAY, 1400

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	37,8	0,01554	60	400	1077	1091
Vent - Return Mixing	Outlet	29,5	0,01209	136	778	-	-
Central Cooling Coil	Outlet	13,1	0,00838	136	778	2693	1487
Supply Fan	Outlet	13,5	0,00838	136	778	76	-
Cold Supply Duct	Outlet	13,8	0,00838	133	778	-	-
Zone Air	-	22,7	0,00939	133	1082	1433	396
Return Plenum	Outlet	23,1	0,00939	133	1082	65	-
Duct Leakage Air	Outlet	13,5	0,00839	3	778	-	-
Return Duct	Outlet	22,9	0,00937	136	1076	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1427	Cooling	1433	22,7	133	1082	0	0

Air System Information

Air System Name	AC-09	Number of zones	1
Equipment Class	SPLT AHU	Floor Area	7,8 m²
Air System Type	SZCAV	Location	Rio de Janeiro, Brazil

Sizing Calculation Information

Calculation Months	Jan to Dec		
Sizing Data	Calculated	Zone L/s Sizing	Sum of space airflow rates
		Space L/s Sizing	Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load	2,8	kW	Load occurs at	Aug 1500
Sensible coil load	2,2	kW	OA DB / WB	38,1 / 25,6 °C
Coil L/s at Aug 1500	168	L/s	Entering DB / WB	24,8 / 18,1 °C
Max block L/s	168	L/s	Leaving DB / WB	14,0 / 13,1 °C
Sum of peak zone L/s	168	L/s	Coil ADP	12,1 °C
Sensible heat ratio	0,783		Bypass Factor	0,150
m²/kW	2,8		Resulting RH	55 %
W/m²	357,7		Design supply temp.	13,0 °C
Water flow @ 5,6 °K rise	N/A		Zone T-stat Check	1 of 1 OK
			Max zone temperature deviation	0,0 °K

Supply Fan Sizing Data

Actual max L/s	168	L/s	Fan motor BHP	0,12	BHP
Standard L/s	168	L/s	Fan motor kW	0,09	kW
Actual max L/(s-m²)	21,54	L/(s-m²)	Fan static	300	Pa

Outdoor Ventilation Air Data

Design airflow L/s	23	L/s	L/s/person	5,63	L/s/person
L/(s-m²)	2,88	L/(s-m²)			

Zone Sizing Summary for AC-09

Air System Information

Air System Name **AC-09**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **7,8** m²
Location **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,8	165	165	Dec 1200	0,2	7,8	21,11

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Reunião 2	1	1,8	Dec 1200	165	0,2	7,8	21,11

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Aug 1500 COOLING OA DB / WB 38,1 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	4 m²	377	-	4 m²	-	-
Wall Transmission	4 m²	204	-	4 m²	49	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	4 m²	301	-	4 m²	105	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	94 W	51	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	369 W	332	-	0	0	-
People	4	202	180	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	147	18	10%	15	0
>> Total Zone Loads	**-**	**1614**	**198**	**-**	**170**	**0**
Zone Conditioning	-	1597	198	-	50	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	28	-	0	0	-
Return Fan Load	168 L/s	0	-	168 L/s	0	-
Ventilation Load	23 L/s	418	407	23 L/s	33	0
Supply Fan Load	168 L/s	93	-	168 L/s	-93	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	48	-	3%	5	-
>> Total System Loads	**-**	**2185**	**605**	**-**	**-5**	**0**
Central Cooling Coil	-	2185	605	-	0	0
>> Total Conditioning	**-**	**2185**	**605**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

August DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	38,1	0,01554	23	400	418	407
Vent - Return Mixing	Outlet	24,8	0,01023	168	1096	-	-
Central Cooling Coil	Outlet	14,0	0,00900	168	1096	2185	605
Supply Fan	Outlet	14,5	0,00900	168	1096	93	-
Cold Supply Duct	Outlet	14,7	0,00900	165	1096	-	-
Zone Air	-	22,7	0,00941	165	1206	1597	198
Return Plenum	Outlet	22,9	0,00941	165	1206	28	-
Duct Leakage Air	Outlet	14,5	0,00900	3	1096	-	-
Return Duct	Outlet	22,7	0,00940	168	1204	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1614	Cooling	1597	22,7	165	1206	0	0

Air System Information

Air System Name	**AC-10**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**6,3** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2,4**	kW
Sensible coil load	**1,6**	kW
Coil L/s at Aug 1500	**96**	L/s
Max block L/s	**96**	L/s
Sum of peak zone L/s	**96**	L/s
Sensible heat ratio	**0,688**	
m²/kW	**2,6**	
W/m²	**377,4**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Aug 1500**	
OA DB / WB	**38,1 / 25,6**	°C
Entering DB / WB	**27,7 / 19,9**	°C
Leaving DB / WB	**13,6 / 12,6**	°C
Coil ADP	**11,1**	°C
Bypass Factor	**0,150**	
Resulting RH	**54**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**96**	L/s
Standard L/s	**96**	L/s
Actual max L/(s-m²)	**15,29**	L/(s-m²)
Fan motor BHP	**0,07**	BHP
Fan motor kW	**0,05**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**30**	L/s
L/(s-m²)	**4,76**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-10

Air System Information

Air System Name **AC-10**
Equipment Class ..... **SPLT AHU**
Air System Type ..... **SZCAV**

Number of zones **1**
Floor Area ..... **6,3** m²
Location ..... **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months ..... **Jan to Dec**
Sizing Data ..... **Calculated**

Zone L/s Sizing ..... **Sum of space airflow rates**
Space L/s Sizing ..... **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,0	94	94	Oct 1800	0,1	6,3	14,99

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Reunião 3	1	1,0	Oct 1800	94	0,1	6,3	14,99

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Aug 1500 COOLING OA DB / WB 38,1 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	6 m²	245	-	6 m²	74	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	76 W	41	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	472 W	425	-	0	0	-
People	4	202	180	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	91	18	10%	7	0
>> Total Zone Loads	**-**	**1004**	**198**	**-**	**81**	**0**
Zone Conditioning	-	983	198	-	18	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	23	-	0	0	-
Return Fan Load	96 L/s	0	-	96 L/s	0	-
Ventilation Load	30 L/s	547	543	30 L/s	31	0
Supply Fan Load	96 L/s	54	-	96 L/s	-54	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	30	-	3%	2	-
>> Total System Loads	**-**	**1636**	**741**	**-**	**-2**	**0**
Central Cooling Coil	-	1636	741	-	0	0
>> Total Conditioning	**-**	**1636**	**741**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

August DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	38,1	0,01554	30	400	547	543
Vent - Return Mixing	Outlet	27,7	0,01131	96	836	-	-
Central Cooling Coil	Outlet	13,6	0,00870	96	836	1636	741
Supply Fan	Outlet	14,1	0,00870	96	836	54	-
Cold Supply Duct	Outlet	14,3	0,00870	94	836	-	-
Zone Air	-	23,0	0,00941	94	1037	983	198
Return Plenum	Outlet	23,2	0,00941	94	1037	23	-
Duct Leakage Air	Outlet	14,1	0,00870	2	836	-	-
Return Duct	Outlet	23,0	0,00940	96	1033	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1004	Cooling	983	23,0	94	1037	0	0

Air System Information

Air System Name	**AC-11**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**73,9** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**18,5**	kW
Sensible coil load	**13,3**	kW
Coil L/s at Jul 1500	**1010**	L/s
Max block L/s	**1010**	L/s
Sum of peak zone L/s	**1010**	L/s
Sensible heat ratio	**0,715**	
m²/kW	**4,0**	
W/m²	**250,7**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Jul 1500**	
OA DB / WB	**38,1 / 25,6**	°C
Entering DB / WB	**24,1 / 18,0**	°C
Leaving DB / WB	**13,2 / 12,4**	°C
Coil ADP	**11,3**	°C
Bypass Factor	**0,150**	
Resulting RH	**58**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**1010**	L/s
Standard L/s	**1009**	L/s
Actual max L/(s-m²)	**13,66**	L/(s-m²)
Fan motor BHP	**0,71**	BHP
Fan motor kW	**0,56**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**90**	L/s
L/(s-m²)	**1,22**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-11

Air System Information

Air System Name **AC-11**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **73,9** m²
Location **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	10,7	989	989	Jul 1600	0,7	73,9	13,39

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Staff 1	1	10,7	Jul 1600	989	0,7	73,9	13,39

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 38,1 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	14 m²	1980	-	14 m²	-	-
Wall Transmission	23 m²	871	-	23 m²	269	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	14 m²	1084	-	14 m²	378	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	887 W	482	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	1356 W	1221	-	0	0	-
People	12	606	540	0	0	0
Infiltration	-	3399	2894	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	964	343	10%	65	0
>> Total Zone Loads	**-**	**10607**	**3777**	**-**	**712**	**0**
Zone Conditioning	-	10436	3777	-	148	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	266	-	0	0	-
Return Fan Load	1010 L/s	0	-	1010 L/s	0	-
Ventilation Load	90 L/s	1671	1495	90 L/s	368	0
Supply Fan Load	1010 L/s	561	-	1010 L/s	-561	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	318	-	3%	21	-
>> Total System Loads	**-**	**13252**	**5272**	**-**	**-23**	**0**
Central Cooling Coil	-	13252	5274	-	0	0
>> Total Conditioning	**-**	**13252**	**5274**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	38,1	0,01554	90	400	1671	1495
Vent - Return Mixing	Outlet	24,1	0,01041	1010	637	-	-
Central Cooling Coil	Outlet	13,2	0,00863	1010	637	13252	5274
Supply Fan	Outlet	13,7	0,00863	1010	637	561	-
Cold Supply Duct	Outlet	13,9	0,00863	989	637	-	-
Zone Air	-	22,7	0,00993	989	660	10436	3777
Return Plenum	Outlet	22,9	0,00993	989	660	266	-
Duct Leakage Air	Outlet	13,7	0,00864	20	637	-	-
Return Duct	Outlet	22,7	0,00991	1010	660	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	10607	Cooling	10436	22,7	989	660	0	0

Air System Information

Air System Name	**AC-12**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**9,2** m²
Location	**Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**3,9**	kW
Sensible coil load	**3,1**	kW
Coil L/s at Jul 1500	**227**	L/s
Max block L/s	**227**	L/s
Sum of peak zone L/s	**227**	L/s
Sensible heat ratio	**0,791**	
m²/kW	**2,4**	
W/m²	**423,7**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Jul 1500**	
OA DB / WB	**38,1 / 25,6**	°C
Entering DB / WB	**24,8 / 17,9**	°C
Leaving DB / WB	**13,6 / 12,6**	°C
Coil ADP	**11,6**	°C
Bypass Factor	**0,150**	
Resulting RH	**52**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**227**	L/s
Standard L/s	**227**	L/s
Actual max L/(s-m²)	**24,65**	L/(s-m²)
Fan motor BHP	**0,16**	BHP
Fan motor kW	**0,13**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**30**	L/s
L/(s-m²)	**3,26**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-12

Air System Information

Air System Name **AC-12**
Equipment Class ..... **SPLT AHU**
Air System Type ..... **SZCAV**

Number of zones **1**
Floor Area ..... **9,2** m²
Location ..... **Rio de Janeiro, Brazil**

Sizing Calculation Information

Calculation Months ..... **Jan to Dec**
Sizing Data ..... **Calculated**

Zone L/s Sizing ..... **Sum of space airflow rates**
Space L/s Sizing ..... **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	2,4	222	222	Oct 1500	0,1	9,2	24,15

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Copa 1	1	2,4	Oct 1500	222	0,1	9,2	24,15

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 38,1 °C / 25,6 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 15,0 °C / 9,7 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	2 m²	130	-	2 m²	-	-
Wall Transmission	4 m²	185	-	4 m²	49	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	2 m²	120	-	2 m²	42	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	110 W	60	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	1550 W	1396	-	0	0	-
People	4	216	220	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	211	22	10%	9	0
>> Total Zone Loads	**-**	**2318**	**242**	**-**	**100**	**0**
Zone Conditioning	-	2301	242	-	43	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	33	-	0	0	-
Return Fan Load	227 L/s	0	-	227 L/s	0	-
Ventilation Load	30 L/s	553	573	30 L/s	75	0
Supply Fan Load	227 L/s	126	-	227 L/s	-126	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	70	-	3%	3	-
>> Total System Loads	**-**	**3083**	**815**	**-**	**-5**	**0**
Central Cooling Coil	-	3083	815	-	0	0
>> Total Conditioning	**-**	**3083**	**815**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	38,1	0,01554	30	400	553	573
Vent - Return Mixing	Outlet	24,8	0,00992	227	986	-	-
Central Cooling Coil	Outlet	13,6	0,00870	227	986	3083	815
Supply Fan	Outlet	14,0	0,00870	227	986	126	-
Cold Supply Duct	Outlet	14,3	0,00870	222	986	-	-
Zone Air	-	22,9	0,00907	222	1078	2301	242
Return Plenum	Outlet	23,0	0,00907	222	1078	33	-
Duct Leakage Air	Outlet	14,0	0,00870	5	986	-	-
Return Duct	Outlet	22,8	0,00906	227	1076	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,206 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2945,5 W/(L/s) Site Altitude = 6,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	2318	Cooling	2301	22,9	222	1078	0	0